

YIGIT MANAFI

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EDUCATION

Sabanci University, Istanbul — B.Sc. Computer Science and Engineering (Physics Minor)

GPA 3.84 / 4.00 | Expected Graduation: 2028 | Double Major: Industrial Engineering

- High Honour List — 2024–2025 and 2025–2026 academic years.
- Academic Achievement Scholarship (Üstün Başarı Bursu) — 2024–2025 and 2025–2026 academic years.

PROFILE

Computer Science and Engineering student working at the intersection of algorithms, high-performance computing, and software development. Experience in C++ and Python development on Linux-based systems, MPI/OpenMP-based parallel processing of large-scale sparse data, randomized estimation methods, and experimental performance analysis. Co-author of an algorithm engineering paper on sparse-matrix permanent estimation currently under review at ALENEX (SIAM). Additional background in data analysis, mathematical modeling, and hardware-adjacent domains including semiconductor fabrication and digital design. Interested in building reliable, high-performance software for large-scale data systems.

PUBLICATIONS

“The Quick and the Dead: Estimating Sparse-Matrix Permanents with Adaptive Work Filtering”

Under review — ALENEX (SIAM Symposium on Algorithm Engineering and Experiments), 2026 | Sabanci University Parallel Computing Group (Assoc. Prof. Kamer Kaya)

- Co-author of the paper.

Survey on Sparse Matrix Ordering Methods

In preparation — started 10/2025 | Sabanci University Parallel Computing Group (Assoc. Prof. Kamer Kaya)

- Co-author of a literature survey on sparse matrix ordering methods, with a focus on performance analysis and structural evaluation.

RESEARCH

Simula Research Laboratory, Oslo, Norway — Research Intern, High-Performance Sparse Matrix I/O 07/2026 – present

Supervisor: Assoc. Prof. Johannes Langguth (Senior Research Scientist, Simula Research Laboratory & University of Bergen)

- Main goal of the project: build a fast, reusable file read/write system that practitioners can use for sparse matrix workloads, with ongoing work to push the achievable speedup further.
- Developing C++ HPC pipelines for high-throughput binary I/O of very large sparse matrices, with individual matrices around 400 GB in size.
- Minimizing end-to-end time from storage to a usable in-memory CSR representation, and keeping that representation fast to operate on.
- Parallelizing binary file reading and CSR construction with MPI (across nodes) and OpenMP (within nodes) on the ex3 supercomputer (Experimental Infrastructure for Exploration of Exascale Computing, Simula).
- Targeting high-performance SpMV (sparse matrix–vector multiplication) on the loaded matrices.
- Benchmarking I/O throughput, memory bandwidth, and parallel scaling to locate bottlenecks along the storage-to-compute path.

Sabanci University — Research Assistant, Computer Science and Engineering (Sparse Data & Parallel Computing) 08/2025 – present

Supervisor: Assoc. Prof. Kamer Kaya (Parallel Computing)

- Develop C++ features for SparseViz, a Linux-based sparse matrix visualization and performance analysis tool.
- Implemented structural and statistical matrix quality metrics: bandwidth, row/column span, envelope, block density, Jaccard similarity, skewness of nonzero distribution, and row block efficiency.
- Added row–column comparison metrics (Envelope R-C, Jaccard Row-Col) to quantify structural asymmetry after reordering.
- Built HTML-based reporting to compare multiple matrices and ordering methods in a readable, reproducible format.
- Conducted experimental testing across sparse matrix collections to validate metric correctness and identify performance/locality patterns.

Sabanci University — Research Course, Quantum Mechanics: Negative Probabilities 08/2024 – 01/2025

Supervisor: Prof. Zafer Gedik (Quantum Mechanics)

- Researched mathematical interpretations and real-world examples of negative probabilities in quantum mechanics.
- Developed a program to compute and illustrate negative-probability examples.
- Applied mathematical modeling and programming to explore computational cases.

Sabancı University — Research Assistant, Materials Science and Nanoengineering (Ceramics) 02/2024 – 07/2024

Supervisor: Prof. Ozge Akbulut (Harvard, Sabanci University)

- Assisted in experimental research on ceramics, materials engineering, and nanotechnology applications.
- Contributed to data collection, sample preparation, and analysis for laboratory studies.
- Gained hands-on experience with laboratory equipment including CNC and SEM.

PROJECTS

Sparse-Matrix Permanent Estimation — C++ / Linux

- Implemented and benchmarked randomized estimators for matrix permanents, including minimum-degree vertex ordering, matrix scaling, and support filtering.
- Built the experimental harness for multi-run statistical evaluation of estimator accuracy and runtime.

SparseViz Tool Development — C++ / Linux / HTML

- Developed and tested C++ analysis features for sparse matrix visualization and ordering evaluation.
- Built HTML report outputs comparing matrix metrics across ordering methods.
- Worked in a remote Linux server environment with Git/GitHub and an iterative debugging workflow.

Time-Series Data Analysis and Prediction — Python / Jupyter Notebook

- Analyzed 398 days of behavioral and physiological time-series data (sleep, heart rate, activity, calendar variables, mood scores).
- Performed exploratory data analysis and hypothesis testing to identify predictive patterns.
- Extending the project with machine learning models for next-day mood prediction and feature importance analysis.

Selected code samples and further project details are available on my GitHub profile.

EXPERIENCE

NATO STO IST-TSI-244 — Summer School on Dual-Use Quantum Technologies, Participant 07/2026 – 08/2026

- Completed an international NATO Science and Technology Organization training school on dual-use applications of quantum technologies.
- Covered quantum computing, quantum communication, and quantum sensing, together with their civilian and defence-oriented use cases.

TÜBİTAK BİLGEM YİTAL, Kocaeli — Processor Manufacturing Intern 01/2025 – 02/2025

- Completed an internship focused on semiconductor fabrication and processor manufacturing processes.
- Hands-on exposure to photolithography, ion implantation, and integrated circuit production steps.
- Learned how fabrication constraints relate to processor design and computing systems.

Sabancı University — Teaching Assistant, Data Science and Python, Summer High School Program 07/2026 – 09/2026

- Assisted in Data Science and Python courses for high school students.
- Guided students through data analysis workflows, coding exercises, and project work.

Sabancı University — Teaching Assistant, Summer High School Program 07/2025 – 09/2025

Instructors: Assoc. Prof. Kamer Kaya, Yalçın Can Kılıç, Prof. Zafer Gedik, Dr. Alhun Aydın, Dr. Göktuğ Karpat

- Assisted in Introduction to Python and Quantum Physics courses for high school students.
- Supported students in programming, scientific discussion, and problem-solving sessions.

Sabancı University Academic Support Program, Istanbul — Peer Moderator 10/2024 – 08/2025

- Peer moderator for Natural Sciences 101/102 and Math 101/102 (Calculus I & II).
- Reteaching course content after lectures and moderating peer study sessions.

Toastmasters, Kocaeli — Leadership Program 01/2026 – 02/2026

- Developed communication, teamwork, and public speaking skills through structured leadership activities.

QSB SU — Sabanci Quantum Society, Istanbul — General Secretary 07/2024 – present

SKILLS AND LANGUAGES

- **Programming:** C++ (upper intermediate), Python (intermediate), HTML (beginner)
- **Software Development:** Git, GitHub, debugging, modular code design, code review, data processing
- **Algorithms & Computation:** Sparse matrices, permanent estimation, sequential importance sampling, matrix scaling (Sinkhorn–Knopp), Dulmage–Mendelsohn decomposition, minimum-degree ordering, bipartite matching, graph algorithms, algorithm engineering and validation
- **Data & Evaluation:** Jupyter Notebook, time-series analysis, exploratory data analysis, hypothesis testing, statistical variance/error analysis, machine learning basics, experimental design and benchmarking
- **HPC & Parallelism:** MPI, OpenMP, multi-threaded C++, supercomputer workflows (eX3, Simula), binary I/O, CSR formats, SpMV, scaling and bottleneck analysis
- **Systems & Tools:** Linux, SSH, remote server development, VS Code, CMake basics
- **Writing & Reporting:** LaTeX / Overleaf, academic writing, HTML report generation
- **Hardware-Related:** Verilog, LTspice, basic electronics, semiconductor fabrication exposure
- **Languages:** Turkish (Native), English (C1), Turkish Sign Language (Beginner)